Take-Home Project

Title Transfer Agent

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1. Why we're asking you to build this

We manage more than 100 properties and several thousand tenants, plus a full accounting division and an in-house construction team. With so many properties and departments, smaller more menial tasks have become inefficient to complete manually.

The purpose of this take home project is to see how you think about agent design, document intelligence, workflow orchestration, and integration using a real problem we deal with constantly: transferring titles on manufactured and mobile homes between a seller and a buyer.

There is no single right answer. We care more about how you reason, design, and build than about checking off every requirement.

2. The project: an HCD Title Transfer Agent

Build a working prototype of an internal application that lets a user scan an existing manufactured/mobile home title, automatically extract the data from that scan, and export it cleanly into the correct HCD transfer forms producing a signature-ready packet with AI agents doing the heavy lifting and a human approving at the right checkpoints.

Scope: straight buyer-to-seller transactions only

For this take-home, restrict the flow to a standard buyer-seller change of ownership. Do not attempt to handle salvage title transfers or warehouseman's lien transfers those are separate workflows with different forms and legal requirements, and we will evaluate those use cases separately later.

A typical HCD title transfer in our world looks like this:

1. An existing title (issued by the California Department of Housing and Community Development) identifies the current registered owner(s), legal owner/lienholder if any, decal/serial numbers, year, make, and the park/site location.
2. A buyer and seller agree to a sale. We collect the existing title, a bill of sale, and the tax clearance certificate from the county.
3. The seller must release interest, and the buyer must be recorded as the new registered owner which requires the correct set of HCD forms, properly filled in, signed, and submitted.
4. Once the packet is complete, it gets routed internally for review, then submitted to HCD. Downstream, our accounting, tenant, and property systems need to reflect the new owner.

3. The HCD forms you'll need

All official forms come from the California HCD Registration and Titling program. We'll hand you the specific forms to build against alongside the sample title so there's no ambiguity. For this exercise, you'll be targeting:

- **HCD 476.6G — Multi-Purpose Transfer Form:** the primary form used to record a change of ownership on a manufactured or mobile home.

  https://www.hcd.ca.gov/sites/default/files/docs/manufactured-and-mobilehomes/hcd-rt-476-6g.pdf
- **HCD 476.6 — Statement of Facts:** used to declare facts supporting the transfer.

  https://www.hcd.ca.gov/sites/default/files/docs/manufactured-and-mobilehomes/hcd-rt-476-6.pdf
- **HCD 480.5 — Application for Registration:** documents the transfer of ownership.

  https://www.hcd.ca.gov/sites/default/files/docs/manufactured-and-mobilehomes/hcd-rt-480-5.pdf

You can reference the live versions on the HCD website (hcd.ca.gov → Manufactured and Mobile homes → Registration and Titling → Forms & Publications) if you want to double-check anything.

As you will see there is information on the forms that cannot be found on the title and those fields and data will always be the same so those can be set to auto fill.

4. What you should build a rough V1

This is intentionally a short exercise. We want to see a rough V1 you can put together in about an hour not a polished product. Use AI tools aggressively. The point is to see how you think, not how many features you can ship.

Minimum to show us

- **Scan the title:** let us upload the sample title PDF (see §5) and extract the key fields (decal, serial, year/make, registered owner, site address) automatically.
- **Export into HCD forms:** take those extracted fields, plus basic buyer info (you can hard-code or use a simple input), and push them into the correct HCD transfer form(s). A filled PDF or a form preview is fine.
- **Show the flow:** any basic UI or script that demonstrates scan → extract → export end-to-end on the one example we provided.

Explicitly out of scope for V1

- Polished UI, auth, persistence, audit logs.
- Handling salvage title or warehouseman's lien transfers.
- Multi-agent orchestration frameworks, integration write-backs, notifications.
- Anything that can't be cut without losing the scan → form export demonstration.

Human in the Loop (HITL) Aspect

- Show us where or how a human would 'verify' the AI's work before the final PDF is generated.

5. What we'll provide

- One example HCD title to build against. We want to see the scan-to-forms pipeline working cleanly on this real example; extensibility to other titles should be clear from your design even though we're only providing one sample.
- The three HCD forms listed as blank PDFs, so you can focus on extraction and mapping rather than hunting down forms. Examples of how the completed forms should look (highlighted sections)
- Buyer and seller information for that transfer (names, addresses, sale price, date) is supplied separately either via your UI or as a small JSON stub you define.
- A short call (15 minutes) to answer questions if you'd like..

6. Tech choices

Your call. Pick the stack you'd actually pitch us on Day 1 of the job. We'd like you to justify your choices briefly in the README model selection, agent framework, data store, hosting, etc. We have no religion about Python vs. TypeScript or any specific framework. We do care that you can defend the choice.

Use any frontier model (Claude, GPT, Gemini). If cost matters to your design, say so.

- **We would like to see a reasoning explanation here as to *why* you chose a specific tool. Site specific examples to your choice of stack.**
- **Also, please specifically mention one scaling limitation of your chosen stack that could be encountered when attempting to improve AI models solely by increasing their size.**

7. Evaluation

We'll evaluate your submission on the following:

Criterion	What we're looking for	Weight
Scan → Extract	Does the title actually get read and parsed? Are the right fields captured from the sample title we provided?	35%
Export to HCD Forms	Do extracted fields land in the correct HCD form fields? Is the output recognizable as a real HCD packet?	30%
Thinking & Trade-offs	In a short readme or a 2-minute walkthrough: why this approach, what you'd harden if given more time.	20%
How You Used AI Tools	Show us your workflow. We're hiring someone who will build with AI daily tell us what you prompted and how.	15%

8. Suggested time about one hour

We're expecting a rough V1, not a finished product. If you find yourself over an hour, stop and show us what you have. We'd rather see a tight 60-minute slice than a polished project that hides how you make trade-offs.

Phase	Suggested time	What to do
Setup	~10 min	Read the brief, pull the HCD forms you need from hcd.ca.gov, open the sample title.
Build	~40 min	Scan → extract key fields from the title. Map them into the HCD form(s). Get it working end-to-end, even roughly.
Wrap up	~10 min	A short README (or a quick 2-minute screen recording) covering what you did, what you'd do next, and how you used AI tools.

9. What to submit

5. Your code (GitHub repo, a zip, or a shared notebook whatever's fastest for you).
6. The filled HCD form(s) it produced from the sample title.
7. A few sentences or a 2-minute screen recording explaining your approach, what's rough, and what you'd do next.

10. How we'll use this in the interview

In the next round, you'll walk us through your work live with our leadership team. Be ready to:

- Demo the scan → form export on the sample title.
- Talk through how this same idea would extend to work orders, payroll, materials, and tenant management.
- Tell us what you'd harden, automate, or redesign if we gave you another week.

11. Ground rules

- Use AI tools freely (Cursor, Claude, Copilot, etc.). We do too. Just be ready to explain any code you submit.
- If something is ambiguous, make a reasonable assumption, jot it down, and keep moving. We value judgment over completeness.
- Don't over-engineer. A rough V1 that demonstrates the core idea beats a half-finished ambitious build.

Questions? Reply to the email this brief came with. Good luck — we're excited to see what you build.